Design1 Interaction Design交互设计(communicate and interact)

1.Goals -Develop usable products {Usability means easy to learn, effective to use, and provides an enjoyable experience.} - Involve users in the design process

2. Characteristics ID核心特征-➔Users should be involved throughout the development of the project 用户参与整个项目开发➔ Specific usability and user experience goals need to be identified识别, clearly documented, and agreed一致 to at the beginning of the project➔Iteration is needed through the core activities

【➔ Interaction design is the umbrella term covering all of these aspects: → Fundamental to all disciplines, fields, and approaches concerned with researching and designing computer based systems for people(UI,UX,web design，software design)
多团队合作：➔ Benefits → More ideas and designs generated ➔ Disadvantages → Difficult to communicate and progress forward the designs being create】

3.User-centered approach3 principles -Early focus on users and tasks -Empirical measurement -Iterative design -早期关注用户和任务-经验测量-迭代设计

User Experience用户体验

1.Defining ➔ **How a product behaves and is used by people in the real world➔ Cannot design a user experience − only can design for a user experience➔ Hornbæk and Hertzum (2017)(How users perceive a product) Hassenzahl's (2010) model of the user experience{→ Pragmatic实用主义: how simple, practical, and obvious it is for the user to achieve their goals → Hedonic快感: how evocative and stimulating the interaction is to users }**

2. Understanding users (one size does not fit all/ incorrect assumptions/ Be aware of both people's sensitivities and their capabilities)

Accessibility and Inclusiveness无障碍/包容性➔ **1.Accessibility**: the extent to which an interactive product is accessible by as many people as possible(disabilities残)**2. Inclusiveness:** making products and services that accommodate the widest possible number of people(smartphones=available to everyone regardless of their disability, education, age, or income) **3.Understanding disability** [Sensory impairment感障/Physical impairment/Cognitive]➔Impairment障碍分为：→ Permanent永→ Temporary暂 → Situational情(嘈杂)

Usability and User Experience Goal

usability goals 6个目标

1.Effectiveness有效性 Is the product capable of allowing people to learn, carry out their work efficiently, access the information that they need, or buy the goods that they want?

2. Efficiency效率 Once users have learned how to use a product to carry out their tasks, can they sustain a high level of productivity?

3. Safety. What is the range of errors that are possible using the product, and what measures are there to permit users to recover easily from them?

4. Utility.实用Does the product provide an appropriate set of functions that will enable users to carry out all of their tasks in the way they want to do them?

5. Learnability易学性Is it possible for the user to work out how to use the product by exploring the interface and trying certain actions? How hard will it be to learn the whole set of functions in this way?

6. Memorability. What types of interface support have been provided to 提供哪些类型界面支持帮助用户记住如何执行任务help users remember how to carry out tasks, especially for products and operations they use infrequently

比较Usability goals differ from user experience goals → **Usability** is more objective客观: how useful or productive a system is from its own perspective → **User experience** is more subjective:主观how users experience an interactive product from their own perspective

DesignPrinciples设计原则

1.Visibility可见性{Mobile phone battery display}

2.Feedback{sound,highlighting,animation}

3.Constraints约束{-Restricting the possible actions that can be performed限制可以执行的可能操作-Helps prevent user from selecting incorrect options 防止用户选错-Physical objects can be designed to constrain things }

4. Consistency一致性{ Design interfaces to have similar operations and use similar elements for similar tasks. ➔ The main benefit is that consistent interfaces are easier to learn and use一致的界面更容易学习和使用} Internal and external consistency{ Internal consistency内部一致性将操作设计成在app中具有相同行为 refers to designing operations to behave the same within an application → Difficult to achieve with complex interfaces ➔ External consistency refers to designing operations, interfaces, and so on to be the same across applications and devices外部一致性指在不同应用程序和设备之间 → Very rarely the case, based on different designer's preference}

5. Affordances功能可见性➔ def:Refers to 物体属性人们知道如何使用an attribute of an object that allows people to know how to use it. Interfaces are virtual and do not have affordances like physical objects.interfaces are better conceptualized as 'perceived' 可感知的 affordances

Shneiderman's 8 Golden Rules of ID: 1. Strive for consistency努力保持一致 2. Enable frequent users to use shortcuts 寻求普遍可用性3. Offer informative feedback提供反馈 4. Design dialog to yield closure 设计对话框以产生闭包5. Offer simple error handling 防止错误6. Permit easy traversal of actions 允许逆转7. Support internal locus of control 保持用户控制权8. Reduce short-term memory load 减少短期记忆负荷

Recap➔Interaction design is concerned with designing interactive products to support how people communicate and interact in their everyday and working lives➔It is concerned with how to create quality user experiences for services, devices, and interactive products➔It is multidisciplinary多学科, involving many inputs from wide-reaching disciplines and fields➔优化Optimizing the interaction between users and interactive products requires consideration of a number of interdependent factors, including context of use, types of activity, UX goals, accessibility, cultural differences, and user groups使用环境、活动类型、用户体验目标、可及性、文化差异和用户群体。.➔Design principles, such as feedback and simplicity, are useful heuristics for informing, analyzing, and evaluating aspects of an interactive product.

Design2 The Process of Interaction Design ID过程

[The double diamond of design: Discover, Define, Develop, Deliver]

Understanding the problem space理解问题空间➔ Explore {current user experience? a change needed? this change improve the situation?} ➔ Articulating明确 the problem space → {Team effort/Explore different perspectives/Avoid incorrect assumptions and unsupported claims}

2. Importance of involving users用户参与重要性➔Understanding of users' goals leading to better products➔Expectation management {Realistic expectations/No surprises, no disappointments/ Timely training / Communication, but no hype沟通不炒作} ➔Ownership { Make the users active stakeholders 积极/More likely to forgive or accept problems → Can make a big difference in acceptance and success of product}

3. **Degrees of user involvement用户参与度**➔Member of the design team → Full-time: constant input, but lose touch with users → Part-time: patchy input, and very stressful → Participatory design: involve all stakeholders in the early stages of design➔Face-to-face group or individual activities ➔ Online contributions from thousands of users → Online Feedback Exchange (OFE) systems → Crowdsourcing design ideas → Citizen science ➔ User involvement after product release → Customer review analysis

4. **User-centered approach以用户为中心的方法:**

3 principles

1. Early focus on users and tasks早期关注用户和任务: directly studying cognitive, behavioral, anthropomorphic, and attitudinal characteristics

2. Empirical measurement经验测量: users' reactions and performance to scenarios, manuals, simulations, and prototypes are observed, recorded, and analysed

3. Iterative design迭代设计: when problems are found in user testing, fix them and carry out more tests

Four basic activities of Interaction Design ID基本活动

1. **Discovering requirements**需求 for the interactive product.

2. **Designing alternatives**替代方案that meet those requirements.

3. **Prototyping**原型 the alternative designs so that they can be communicated and assessed.

4. **Evaluating**评估 the product and the user experience it offers throughout the process.

Practical Issues

➔ **Who are the users?{ Users are not always obvious- user is "everybody" → More targeted products are associated with specific roles更有针对性的产品与特定的角色相关联 Stakeholders → The individuals or groups that can influence or be influenced by the success or failure of a project → Larger than the group of direct users → Identifying stakeholders helps identify groups to include in interaction design activities识别涉众有助于识别要包含在交互设计活动中的群体 }**

➔ **What are the users' needs?{~~Users rarely know what is possible~~ Instead: → Explore the problem space → Investigate who are the users → Investigate user activities to see what can be improved → Try out ideas with potential users与潜在用户一起尝试你的想法 Focus on peoples' goals, usability, and user experience goals关注人们的目标、可用性和用户体验目标，而不是期望利益相关者阐明需求}**

➔**How to generate alternative designs?如何产生替代方案?Considering alternatives helps identify better designs考虑备选方案有助于确定更好的设计 {'Flair and creativity': research and synthesis天赋和创造力":研究和综合 → Cross-fertilization of ideas from different perspective不同观点的思想交流 → Users can generate different designs → Product evolution based on changing use → Seek inspiration: similar products and domain, or different products and domain寻找灵感:相似的产品和领域，或不同的产品和领域 ➔Balancing constraints and trade-offs平衡约束和权衡}**

➔ **How to choose among alternatives? {Interaction design focuses on externally-visible and measurable behavior外部可见和可测量的行为(Use of PC v.s. CPU, GPU, memory, etc).➔Technical feasibilit技术可行性➔Evaluation with users and stakeholders → Prototypes, not static documentation, because behaviour is key ➔ A/B Testing: → Online method to inform choice between alternatives → Nontrivial to set appropriate metrics and choose user group sets Quality thresholds → Different stakeholder groups have different quality thresholds (如手机拍照) → Usability and user experience goals lead to relevant criteria}**

➔How to integrate interaction design activities with other lifecycle models?➔Integrating interaction design activities in lifecycle models from other disciplines requires careful planning➔Software development lifecycle models are prominent➔Integrating with agile software development is promising because与敏捷软件开发集成是有希望的: → It incorporates tight iterations → It champions early and regular feedback → It handles emergent requirements → It aims to strike a balance between flexibility and structure

Discovering Requirement发现需求 **1.Purpose**: → Explore the problem space to gain insights → Establish a description of what will be developed捕获需求**2.To avoid miscommunication Requirements activity is the stage where miscommunication occurs most commonly** **3.what:**A statement about an intended product that specifies what it is expected to do or how it will perform Different forms and different levels of abstraction (Atomic requirement shell需求列表 /User stories=As a .., I want.. so that..)**4.Different kinds of requirements**➔ Functional requirements → What the system should do➔ Non-functional requirements → The characteristics (sometimes called constraints)**5. 6种常见需求type:** Functional, data, environment, user, usability, user experience

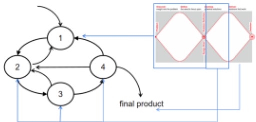
Practical Issues Data gathering for requirements:

➔Interviews, observations, and questionnaires访谈、观察和问卷调查

➔Studying documentation➔ Researching similar products Using probes to engage with users使用探针与用户互动: Many types of probe: → Designed to prompt users into action → For researchers to learn about users ➔ Cultural probe文化调查: → Wallet containing postcards, maps, camera, photo album, and diary → Participants asked to answer questions using wallet contents ➔ Design probe设计调查: → Form relates specifically to particular question and context, for example, Top Trumps probe Contextual Inquiry背景调查➔ Four main principles: → Context: Going to the user, wherever they are, and seeing what they do as they do it → Partnership: User and interviewer explore user's life together → Interpretation: Observations interpreted by user and interviewer together → Focus: Project focus to understand to what should be paid attention Interview in four parts → Overview → Transition → Main interview → Wrap-up ➔ **Persona人物角色** → 对典型用户的丰富描述Rich descriptions of typical users, not specific people Capture a set of user characteristics Develop a small set of personas with one primary (名字岁数背景需求特点目标)

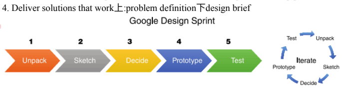
➔ **Scenarios场景** 非正式的叙事性故事→ An informal narrative story, simple, 'natural', personal, and not generalizable➔ Textual descriptions ➔ Animations, audio or video ➔ Example animation scenarios **Use cases** ➔ Focus on functional requirements and capture interaction user stories focuses on outcomes and user goals ➔ Two styles: → Essential use cases: division of tasks, no implementation detail → Use case with normal and alternative courses: more detail (用户使用产品场景)

➔ **use case**描述了系统将如何响应用户的请求以及与用户的交互方式How will the system respond to user requests and interact with users



1.Discover insight-into the problem2.Define-the areatofocus upon 3.Develop potential solutions

4. Deliver solutions that work上:problem definition下design brief



Design3 Visualization **1.DataVisualization**--Computer-based visualization systems provide visual representations of datasets intended to help people carry out some tasks more effectively.

2.Visual representations Visual marks: points, lines and areas. Visual variables: Shape, Size, Value, Orientation, Color, Texture, etc.
3.Representation and Interaction Representation of objects users pay attention to, Interactions which are operations users can apply.
20 Interface types Command/Graphical/Multimedia/Virtual reality/Web/Mobile /Appliance/Voice/Pen/Touch/Gesture/Haptic触觉/Multimodal/Shareable/Tangible有形的/Augmented reality/Wearables/Robots and drones/Brain–computer interaction/Smart

Gesture Interaction: Purpose Requirements: Hygiene卫生: prevent infections or serious complications. Efficiency有效: Simple, direct, no misunderstanding or error.**Touch / Pen-based Interaction** Pros:fast, precise, direct increase the user's impression they are making direct manipulations Cons: limited: used as a discrete interaction mechanism limiting: many complex tasks (in particular for 3D manipulations) require input/control with more than three degrees of freedom**Tangible Interaction** Pros: allow users to achieve complex 3D manipulations with simple real-world style gestures more flexible than other interaction paradigmsManipulations with these devices can be programmed to feel realistic, as they would in the real world Cons: Need extra devices Sometimes, difficult to manipulate**Mid-air gestural interaction** Pros: touchless interaction mimics the physical actions we make in the real world Cons: Not precise Sometimes, difficult to controlHybrid interaction combine multiple interaction paradigms together overcome the inherent limitations of a device augmenting the number of DOF that can be manipulated reduce the occlusion limitation with tactile interaction combine the benefits of two interaction paradigms simply tackle complicated tasks**Interaction types 交互模式**Instructing(教 导)Conversing(对话) Manipulating(操
纵)Exploring(探索)Responding(回应)
Input and output devices of spatial user interface with 3D data 输入输出设备的具有三维数据的空间用户界面
Input(Tactile and pen-based 触觉和基于笔的Tangible and haptic 有形的和触觉的
Mid-air gestural 半空中的手势Hybrid 混合型)**Output**(2D projection: Mobile device
Augmented Reality 增强现实;Virtual Reality) **The Problem: Selection of 3D Subspaces** problem: why/how to efficiently select subspaces in 3D? -raycasting requires large objects -want spatial selection from 2D(Tablet Freehand Lasso) iterative selection too tedious

Prototyping1:

What?- ➔ Instead of an object, it is more a way of thinking. In ID it can be):a series of screen sketches,a storyboard,a PowerPoint slide show.
Emotional design:expressive interfaces,annoying interfaces,anthropomorphism情感设计:富有表现力的界面，恼人的界面，拟人化
Why prototype?(reasons/goals)评估和反馈是ID的核心➔ Stakeholders涉众can see, hold, interact with a prototype more easily than a document or a drawing比文档绘图更易查看互动➔ Team members can communicate effectively ➔ You can test out ideas for yourself ➔ It encourages reflection: very important aspect of design ➔ Prototypes answer questions, and support designers in choosing between alternatives
1.Understanding 避免错误*assumptions*原型帮助理解:**Design alternative**设计替代(多做几种),**Strategy**战略:(Intangible无形→tangible有形) **User-centred processes**流程:1.Don't make false assumptions2.Prototype could build empathy between designers and users with concrete ideas3.Useful tools:Empathy map同理心User journey map
2.Communication increase the efficiency in making decision原型帮助沟通提高效率 Prototype could help you to Communicate with (**Same 'language',Different stakeholder**不同涉众不同处理,**No 'maybe'**原型save efforts in determining style, size, colour, and interactions through observable objects and peer discussion原型可以节省确定风格、大小、颜色以及通过可观察对象和同行讨论进行交互的努力).
3.Test and reflection 测试反思 The main aim of prototype is to examine and improve检验改进 the current design.Prototype could help you to test and reflect on:Hypotheses and assumptions假说和假设(开始设计时,作出假设,如:➔ Users could find the way to specific functions ➔ The information displayed is straightforward ➔ Users could understand the UI and texts ➔ Users' needs could be fully satisfied within this page)
Other's comments其他人的评论:The test and reflection不仅about users, 还within the design group The prototype is an opportunity for you to present and promote the idea to peers, and reflect on the availability and values from different 向同行展示推广
Fidelity:保真度概念The fidelity of a prototype refers to how it conveys the look-and-feel of the final product (basically, its level of detail and realism).Lower fidelity may be useless, 低也许无用higher fidelity may be time- and money-consuming高也许耗时耗钱
Low-fidelity Prototyping低保真(quick cheap easy to change):Uses a medium which is unlike the final medium, e.g., paper, cardboard Is quick, cheap and easily changed
Storyboards ➔ Often used with scenarios, bringing more detail, and a chance to role play ➔ It is a series of sketches showing how a user might progress through a task using the device ➔ Used early in design
Sketching ➔ is important to low-fidelity prototyping Same with the storyboard - Don't be inhibited about drawing ability. Practice simple symbols with pencil and paper
Using index cards (3 X 5 inches) Each card represents one screen Often used in ➔ Website development ➔ Application ➔ MiniProgram
Wizard-of-Oz' prototyping ➔ The user thinks they are interacting with a computer, but a developer is responding to output rather than the system. ➔ Usually done early in design to understand users' expectations ➔ What is 'wrong' with this approach? → It is "Artificial Intelligence"
High-fidelity prototyping高保真: Uses materials that you would expect to be in
the final product. Prototype looks more like the final system than a low-fidelity version.没有明确分界线There is no clear separations
➔ Neither of these is inherently good or bad. ➔ There are costs and trade-offs with raising or lowering fidelity in any dimension.这两种情况本身都不是好或坏。在任何方面提高或降低保真度都有成本和权衡。
Compromises妥协方案:'horizontal': provide a wide range of functions, but with little details广泛功能细节少 'vertical': provide a lot of detail for only a few functions.少数细节大量功能

Prototyping2: Five dimensions in prototyping五维:

Visual: How real does it look?Refers to the devotion and granularity颗粒度put into the design for visual interfaces and physical objects.It is the most direct dimension.投入的精力和颗粒度,最直接的维度 问题- involves jumping to high visual fidelity too early in the process
Interaction: How real does it feel?Refers to the objective degree of exactness with which real-world interactions can be reproduced可再现的准确程度(点击，滑动屏幕)
Breadth: To what degree is this the whole or just a part? it refers to the number of features covered.涵盖的功能数量the entry and exit points to your redesign may matter more than you think.
Depth: At a given level of breadth, to what degree is the user constrained? It refers to the degree of functionality.功能的程度
Content: How real is the stuff and is it contextual to the user?Refers to the degree of realism for the content.真实性程度
Other:Autonomy自主性:operates alone vs requires 'supervision'独自操作与需要监督的关系Platform:interim vs final implementation临时执行与最终执行
Guidelines for physical design实物设计准则:1.Nielsen's heuristics 2. Shneiderman's eight golden rules 3. Styles guides: commercial, corporate decide 'look and feel' for your widgets prescribed, e.g., icons, toolbar
Different kinds of widget不同种类小部件:
menu design:*How long?How many?How will division into groups如何分组➔ What terminology to use? 使用术语How will any physical constraints be accommodated, e.g., mobile phone?适应物理限制*
icon design:*→Meaning of icons is cultural and context sensitive文化和背景敏感度→always draw on existing traditions or standards借鉴现有的→concrete objects or things are easier to represent than actions具体的物体或事物*
Screen design:**1**.moving around within and between screen show much interaction per screen? serial or workbench style?:Splitting functions across screens跨屏幕拆分（Task analysis as a starting point,以任务分析为起点Keep information available保持信息通畅一次性打开多个屏幕）
2.Individual screen design个人屏幕设计,➔ Draw user attention to salient point吸引用户注意Animation is very powerful but can be distracting动画强大但让人分心,Good organization helps: grouping, physical proximity良好组织有助于分组靠近,Trade off between sparse population and overcrowding人口稀少和拥挤的权衡
information display:➔Relevant information available at all times随时提供相关信息
➔Different types of information imply different kinds of display不同信息不同显示方式
➔Consistency between paper display and screen data entry纸张显示和屏幕数据输入之间的一致性
Types:**1.throw-away**扔掉的→ prototype only serves to elicit user reaction → creating prototype must be rapid, otherwise too expensive做原型=引起用户反应，需快否则成本高**2.incremental**递增的→ product built as separate components (modules) → each component prototyped and tested, then added to the final system作为独立组件构建的产品。每个原型化和测试，添加到最后系统 **3.evolutionary**演变→ prototype altered to incorporate design changes → eventually becomes the final product原型变更纳入设计变更做最终产品
Cautions注意事项
➔Rationalize cost-value tradeoffs合理确定成本价值权衡➔Do not oversell - capabilities that can't be delivered, development completeness 不能过度推销-不能交付的，发展的完整性➔Do not overbuild–"good enough"as a prototype 不要过度建设
➔Decide early on exploratory or evolutionary prototypes尽早决定探索性或进化性原型

Ideation	Low fidelity paper sketches
Conceptual design	Low fidelity paper sketches, storyboards
Intermediate design	Low to medium fidelity wireframes
Detailed design	High fidelity wireframes, programmed prototypes

Prototyping3: **:Processes in Prototyping**based on your purpose, audience, and assumptions来选择process.
1.The minimum viable product (MVP)最低限度的可行产品➔ A product with enough features to attract earlyadopter customers and validate a product idea early in the product development cycle. 具有特性的产品，吸引早期用户在产品开发早期验证产品理念➔ There are **4 steps for MVP**: → Understand your users and identify the problem → Write down the user flow → Use the prototype to optimize the user flow → Test, refine, iterate
2.exploration探索➔ Focusing on exploration makes you spent more time in thinking rather than doing. 专注于探索会让你花更多时间在思考而不是行动上。➔ Key points:(Find the right issue to solve → Find solutions → Identify where we are now) ➔ **3 Steps:** → **1.** Brainstorming about issues/solutions头脑风暴(→ Change "No, but" to "Yes, and" → No wrong answers)→ **2.** Clustering and categorization 聚类分类Identify the directions/main issues/topics·识别方向/主要问题/主题 → **3.** Priority check优先检查
Value vs Effort优先级:
High value and low effort – top priority最优先事项
High value and high effort – high risk高风险
Low value and low effort – high utility高效
Low value and high effort – low utility低效
3.Specific audience:**(1)**Identify your audience, purpose, and fidelity **(2)**Tools Tools for different stakeholders → Designer ➔No need for the whole user flow ➔ Specific functions → Sponsor & Customer ➔ Better to provide the whole user flow ➔ A "complete" experience for the product → Developer ➔ Confirmed assumptions with no implementation ➔ Provide details**(3)**Present➔ Make sure you tell a good "story" → Be clear about what you are looking for → Record the feedbacks → Be a bridge between different stakeholders
Different stakeholders不同的利益相关者:
→ Designer设计师 – feedback, support, criticism, idea
·Low fidelity for idea and concept
·High fidelity for details
Tool:·No need for the whole user flow
·Specific functions
→ Sponsor 赞助商–approval, progress, resource
·Plans, progress, expectations
· Like your boss
→ Customer 客户– contract, purchase
·Styles, user flow, compatibility
·Updates
Tools of Sponsor&Customer:·Better to provide the whole user flow
·A "complete" experience for the product
→ Developer 开发人员– feasibility, coding, technical
·Feasibility
·Low fidelity for architecture
·High fidelity for implementation
Tools:·Confirmed assumptions with no implementation
·Provide details
4.Assumptions➔ Focusing on assumptions makes you to improve your ideas and products. The assumptions could be either abstract or concrete.:**4 steps**:(1)Understand your audience, issues, and assumptions**(2)**Fidelity 选择保真度水平For concepts, terminologies, navigation, and user group identification → Low fidelity ➔ For completion, visual details, contents → High fidelity **(3)**Purpose of the testing**(4)**Make and test 只做对你目标有用的，不做用户引导Make prototypes based on your plan → Only make the useful ones for your purpose → E.g., no need to build the UI interface if you just want to make sure that tempo is useful for people to get focused ➔ Testing → Do nothing in guiding the user If they do not act as you expected, just record → Ask questions after the testing → Find new possibilities for your product and iterate
(3)Tools to use:→ Clicks or actions:For user's inputs→ A/B test:To test alternatives → Procedures:To test user flow
Make a plan:1.Purpose and assumptions 2.User groups3.Questions4.Tasks
Digital prototyping: 数字原型1.Screen 2.Responsive design(Mobile-based or Desktop-based)屏幕间的断点(做大中小移动)3.Interaction types-How user inputs?Click,Double-click, pinch,Swipe, Flip 4.Accessibility-Visual Physical,Experience or preference 5.Animation→(Design as earlier as possible To create space for designers to implement → Use drawing for low-fidelity → Use software for high-fidelity)
physical prototyping:物理原型focuses more on the :Electronic technology,Compatibility, Material and tactility电子技术➔兼容性材料和触感➔ Reliability and consistency ➔ Robust and free from errors (connection lost, signal errors)

Evaluation1Heuristic Evaluation and walkthroughs启发式评估

Nielsen's 10 heuristics:1. Visibility of System Status 系统状态可见性 The design should always keep users informed about what is going on, through appropriate feedback within a reasonable amount of time. Eg.Visual and auditory feedback when adjusting the PC sound volume**2.Match Between System and the Real World 系统与现实世界匹配**The design should speak the users' language. Use words, phrases, and concepts familiar to the user, rather than internal jargon. Follow real-world conventions, making information appear in a natural and logical order. Eg.Calculators on your smartphone.**3.User Control and Freedom 用户控制和自由**[Users often perform actions by mistake. controls that allow people to go back to the previous state of the system Eg.Restore a deleted file/Reset a slide layout/Recall a message/email?] **4.Consistency and Standards 一致-遵循平台惯例**Users should not have to wonder whether different words, situations, or actions mean the same thing. Follow platform and industry conventions. Eg.Internal consistency within Microsoft Office ➔ External consistency in ecommerce utility navigation**5. Error Prevention错误诊断**Good error messages are important, but the best designs carefully prevent problems from occurring in the first place.Eg.National Geographic VR showed a confirmation message to protect users from accidentally leaving the activity and losing progress **6. Recognition Rather than Recall识别而不是回忆**[Minimize the user's memory load by making elements, actions, and options visibleEg.Excel sheet without the header ➔ Search history] **7. Flexibility and Efficiency of Use 灵活性/使用效率**Shortcuts hidden from novice users may speed up the interaction for the expert user such that the design can cater to both inexperienced and experienced users. Allow users to tailor frequent actions.Eg.Adjust different window size and layout in Unity. ➔ Blender shortcuts for experienced users**8. Aesthetic and Minimalist design 美学极简设计**Dialogues should not contain information which is irrelevant or rarely needed. Every extra unit of information in a dialogue competes with the relevant units of information and diminishes their relative visibility. Eg.Google's minimalist homepage was tailored to user's primary action – search. Baidu has more information cluttered on the homepage.**9. Help Users Recognize, Diagnose, and Recover from Errors**帮**user识别诊断从错误中恢复**[Error messages should be expressed in plain language (no error codes), precisely indicate the problem, and constructively suggest a solution.] **10.Help and Documentation帮助文档** it may be necessary to provide documentation to help users understand how to complete their tasks.Eg.Learning mall provides a knowledge base to help teachers set up activities.

Doing heuristic evaluation进行启发式评价1. Briefing session to tell experts what to do 2. Evaluation period of 1-2 hours in which: ➙ Each expert works separately ➙ Take one pass to get a feel for the product ➙ Take a second pass to focus on specific features 3. Debriefing session in which experts work together to prioritize problems➔ Suggesting tasks may be helpful, but can be difficult if the evaluation is done early in design when there are only screen mock-ups or a specification. ➔ A second researcher may record the problems identified. ➔ Strategies such as 'think aloud' can be used, and the process can be video recorded.

Advantages and problems➔ Few ethical and practical issues to consider because users are not involved ➔ Can be difficult and expensive to find experts ➔ Best experts have knowledge of application domain and users ➔ Biggest problems(Important problems may get missed /Many trivial problems are often identified, such as false alarms / Experts have biases)A more detailed procedure1. Know what to test and how2. Know your users and have clear definitions of the target audience's goals, contexts, etc. User personas can help evaluators see things from the users' perspectives. 3. Select 3–5 evaluators, ensuring their expertise in usability and the relevant industry. 4. Define the heuristics (around5-10) depend on the nature of the system/product/design. 5. Brief evaluators on what to cover in a selection of tasks, suggesting a scale of severity codes (e.g., critical) to flag issues. 6. 1st Walkthrough – Have evaluators use the product freely so they can identify elements to analyze. 7. 2nd Walkthrough – Evaluators scrutinize individual elements according to the heuristics. They also examine how these fit into the overall design, clearly recording all issues encountered. 8. Debrief evaluators in a session so they can collate results for analysis and suggest fixes

Cognitive walkthroughs认知调查

➔ Simulating how users go about problem-solving at each step in a human-computer interaction模拟用户在人机交互的每个步骤中如何解决问题

➔ Focus on ease of learning步骤:

1. Preparations ➙ Identify and document the characteristics of typical users典型用户的特征. ➙ Develop sample tasks,样例任务 focusing on the aspects of the design. ➙ Produce a description, mock-up, or prototype of the interface to be developed, along with a clear sequence of the actions needed for the users to complete the task用户完成任务所需的清晰操作顺序.
2. A designer and one or more researchers come together to do the analysis.
3. The researchers walk through the action sequences for each task, placing it within the context of a typical scenario. As they do this, they try to answer three questions.
4. Compile a record of critical information – assumptions, notes of issues and design changes, etc.
5. (Check with real users and) Revise the design to fix the problems presented

1. Will the correct action be sufficiently evident to the user?
➙ Users know what to do
2. Will the user notice that the correct action is available?
➙ Users know how to do it
3. Will the user associate and interpret the response from the action correctly?
➙ Users understand the feedback

Questionnaires➔ Questions can be closed-ended or open-ended ➔封闭问题易分析 Closed questions are easier to analyze, and may be distributed and analyzed by computer ➔ They can be administered to large populations ➔ Disseminated by paper, email and the web

Design➔ The impact of a question can be influenced by question order.**受顺序影响** ➔ You may need different versions of the questionnaire for different populations.不同人群不同版本➔ Provide clear instructions on how to complete the questionnaire. ➔ Avoid very long questions and questionnaires.**避免长**➔ Decide on whether phrases will all be positive, all negative, or mixed.**确定问题性质**➔ Strike a balance between using white space and keeping the questionnaire compact.

Question and response format ➔ 'Yes' and 'No' checkboxes ➔ Checkboxes that offer many options ➔ Rating scales量表(Likert scales ➙ Semantic scales ➙ 3, 5, 7 or more points) ➔ Open-ended responses

Encouraging a good response 鼓励好的回应➔ Make sure that the purpose of study is clear ➔ Promise anonymity**承诺匿名**➔ Ensure that questionnaire is well designed ➔ Offer a short version for those who do not have time to complete a long questionnaire ➔ **跟进**Follow-up with emails, phone calls, or letters ➔ Provide an incentive**提供激励**➔ If mailed, include a stamped, addressed envelope

Pros 1.collect data from a large number of people, at a relatively low cost低成本大数据 2.get an overview of a population of users in a short amount of time 短时间3. 不需要any special equipment 4. Surveys are generally approved by institutional review boards because they are typically non-intrusive非侵入性

cons1. good at getting shallow**浅** data from a large number of people, but are not good at getting "deep" data 2. Since surveys are usually self-administered, it is usually not possible to ask follow-up questions 3. Surveys can lead to biased data when the questions are related to patterns of usage, or feelings about a previous experience, rather than clear factual phenomena**可能会有偏见** ➙ Eg.how many times did you use this software application over 6 months?

Deploying online questionnaires1.Plan the timeline 2. Design the questionnaire offline 3. Program/complete online survey **4.** Test the survey to make sure that it behaves as you would expect **5.** Test it with a group that will not be part of the survey to check that the questions are clear 6. Recruit participants

Interviews

Four types:

➔ **Unstructured**: Not directed by a script. Rich but not replicable.非结构化:不受剧本指导。丰富但不可复制

➔**Structured**: Tightly scripted, often like a questionnaire. Replicable but may lack richness. 结构化:脚本严密，通常像问卷。

➔**Semi-structured**: Guided by a script, but interesting issues can be explored in more depth. Can provide a good balance between richness and replicability.半结构化:以剧本为指导，但可以更深入地探讨有趣的问题。可以在丰富性和可复制性之间提供很好的平衡

➔ **Focus groups:** A group interview

questions需avoid： ➔ Long questions ➔ Compound sentences (➙ Split them into two) ➔ Jargon and language that the interviewee may not understand ➔ Leading questions that make assumptions ➙ (eg.Why do you like …?) ➔ Unconscious biases (➙gender stereotypes) [Insufficient rapport不够融洽/not probing for details/multitasking during the interview/allowing observes to influence the interview/leading the participant]

Running the interview1. **Introduction**: Introduce yourself, explain the goals of the interview, reassure about the ethical issues, ask to record, and present the informed consent form. 2. **Warm-up:** Make first questions easy and non-threatening.**简单不具威胁性**➙ Build rapport (a close and harmonious relationship) 3. **Main body:** Present questions in a logical order 4. **A cool-off period:** Include a few easy questions to defuse tension at the end 5. Closure: Thank interviewee, signal the end, for example, switch recorder off. **Digital conferencing systems**(Skype, Zoom, email, and smartphones)➔**Pros** (Go deep: encourage reflection and consideration➙Flexible: open-ended and exploratory) **Cons** (Skill to manage ➙Time and resource intensive ➙ Data analysis ➙ Recall problems Separated from the task and context under consideration)

Choosing and combining techniques➔Depends on the:➙Focus of the study➙Participants involved➙Nature of the technique(s)➙Resources available➙Time available

Five key issues

1. Setting goals➙What information to collect ➙ How to analyze data once collected
2. Identifying participants➙Decide from whom to gather data and how many 识别参与者，决定向谁收集数据及多少数据
3. Relationship with participants➙Clear and professional➙Informed consent when appropriate
4. **Triangulation三角测量➙从多个角度看数据，·收集多种类型的数据Look at data from more than one perspective ➙Collect more than one type of data, for instance, quantitative data from experiments and qualitative data from interviews 5. Pilot studies试点研究**

Summary ➔ Questionnaires may be on paper, online, or telephone调查问卷可以是书面、网上或电话形式 ➔ Interviews may be structured, semi-structured, or unstructured ➔ Focus groups are group interviews ➔ Observation may be direct or indirect, in the field, or in controlled settings ➔ Techniques can be combined depending on the study focus, participants, nature of technique, and available resources and time ➔ Data may be recorded using handwritten notes, audio or video recording, a camera, or any combination of these ➔ Data gathering sessions should have clear goals 数据收集会议应该有明确的目标

Evaluation2Why, what, where and when to evaluate

Iterative design and evaluation is a continuous process.

Why: To **check** users' requirements and confirm that users can utilize the product and that they like it**检查用户需求**➔

What: A conceptual model,**概念模型** early and subsequent prototypes of a new system, more complete prototypes, and a prototype to compare with competitors' products ➔ **Where**: In natural, in-the-wild, and laboratory settings**自然/实验室**➔

When: Throughout design**整个设计过程**; finished products can be evaluated to collect information to inform new products

HCI Research Methods and Measurement Early days HCI research measurements were based on human performance (task-based)**基于人的表现**(Time performance ➙ Task correctness / accuracy ➙ Error rate ➙ Time to learn and retention over time ➙ User satisfaction)

Different types of evaluation method不同评估方法

1. **Controlled settings** that directly involve users直接涉及用户的控制设置 ➙ For example, usability and research labs
2. **Natural settings** involving user涉及用户的自然环境➙ For instance, online communities and products that are used in public places ➙ Often there is **little or no control** over what users do, especially in in-the-wild settings
3. **Any setting that doesn't directly involve users**任何不直接涉及用户的设置➙ For example, consultants and researchers critique the prototypes, and may predict and model how successful they will be when used by users

Method	Controlled settings	Natural settings	Without users
Examples	✓Usability testing ✓Experimental design	✓Field study	Heuristic evaluation/ Analytics/A/B testing Predictive models

Practical challenges of evaluation评估的实际挑战

1.get consent获得同意➔ Participants need to be told why the evaluation is being done, what they will be asked to do and informed about their rights ➔ **Informed consent forms知情同意书**provide this information and act as a contract between participants and researchers ➔ The design of the informed consent form, the evaluation process, data analysis, and data storage methods are typically approved by a high authority, such as the Institutional Review Board

2.Interpreting data解释数据➔ **Reliability**可靠性: Does the method produce the same results相同结果on separate occasions不同场景? ➔ **Validity**有效性: Does the method measure what it is intended to measure? ➔ **Ecological validity**生态有效性: 评估环境是否distort扭曲结果➔ **Biases:** Are there biases that distort the results? ➔ **Scope**范围:结果有多普遍generalizable?

Usability testing➔ Controlled settings ➔ Users are observed and timed观察用户并计时➔ Data is recorded on video, and key presses are logged ➔ The data is used to calculate performance times and to identify and explain errors ➔ User satisfaction is evaluated using questionnaires and interviews使用问卷和访谈评估用户满意度➔ Field observations may be used to provide contextual understanding实地观察

➔ Involves recording performance of typical users doing typical tasks记录典型用户执行典型任务的性能

Quantitative performance measures➔ Number of users successfully completing the task ➔ Time to complete task ➔ Time to complete task after time away from task ➔ Number and type of errors per task ➔ Number of errors per unit of time ➔ Number of navigations to online help or manuals ➔ Number of users making a particular type of error ➔ Count and calculate data **conditions条件**➔ Usability lab or other controlled space ➔ Emphasis on: ➙ Selecting **representative users** ➙ Developing **representative tasks** ➔ 5-10 users typically selected ➔ Tasks usually around 30 minutes ➔ **Test conditions are the same** for every participant ➔ **Informed consent form** explains procedures and deals with ethical issues**多少人够?** ➔ Depends on:➙Schedule for testing ➙Availability of participants ➙ Cost of running tests ➔ **Typically 5-10 participants ➔continue until no new insights are gained Usability testing and Experiments**➔ Usability testing is applied experimentation ➔ Developers check that the system is **usable** by the intended user population by collecting data about **participants'performance** on prescribed **tasks**收集数据检查➔ Experiments test **hypotheses** 假设to discover new knowledge by investigating the **relationship** between two or more **variables**两个或多个变量之间关系Usability Testing (Improve products ➔ Few participants➔Results inform design➔Usually not completely replicable➔ Conditions controlled as much as possible➔Procedure planned➔Results reported to developers)Experiments for Research➔ Discover knowledge ➔ Many participants ➔ Results validated statistically ➔ Must be replicable ➔ Strongly controlled conditions ➔ Experimental design ➔ Scientific report to scientific community

Experimental design实验设计 Experiments➔ Test hypothesis测试假设➔Predict the relationship between two or more variables两个及多个变量➔Independent自变量variable is manipulated by the researcher ➔ Dependent variable因变量influenced by the independent variable ➔ Typical experimental designs have one or two independent variables ➔ Validated statistically and replicable统计验证和可复制

Research Hypotheses➔ An experiment normally starts with a research hypothesis ➔ A hypothesis is a precise problem statement 假设是一种精准问题陈述that can be directly tested through an empirical investigation

Types of Hypothese假设类型➔ Null hypothesis: typically states that there is **no difference** between experimental treatments ➔ **Alternative hypothesis**: a statement that is mutually exclusive with the null hypothesis ➔ The goal of an experiment is to find statistical evidence to reject the null hypothesis in order to support the alternative hypothesis实验目标是找到统计证据拒绝零假设来支持替代假设

Experimental designs Between subjects design(Different participant不同参与者➙ Single group of participants is allocated randomly to the experimental conditions优点:No order effects缺点:Many subjects and individual differences is a problem**)Within subjects design (**Same participants同一参与者➙All participants appear in both conditions优点:Few individuals, no individual differences缺点:Counter-balancing needed because of ordering effects**)Good Practices** Counterbalancing condition orders through a Latin Square Design通过拉丁方设计平衡条件订单.Provide sufficient training time for users to get acquainted with a system to eliminate the learning effect(<60-90min)

Experimental design lifecycles实验设计生命周期 1. Identify a research **hypothesis** 2. Specify the **design** of the study 3. Run a **pilot study** to test the design, the system, and the study instruments 4. Recruit participants 5. Run the **actual data collection sessions** 6. Analyze the data 7. Report the results

Experiment procedure (Step 5)实验步骤1. Preparation 2. Greet participants 3. Introduce the purpose of the study and the procedures 4. Get consent 5. Assign participants to a specific experiment condition 6. Training task(s) 7. Actual task(s) 8. Participants answer questionnaires (if any) 9. Debriefing session 10. Payment (if any)

Statistical analysis

Given the sample data: {5, 5, 4, 3, 4, 1, 5, 4, 4, 4}

➔ What is the sample size?

⇾ =COUNT(B2:B11)

➔ What is the mean value?

⇾ =AVERAGE(B2:B11)

➔ What is the standard deviation?

⇾ =STDEV.S(B2:B11)

Given another sample data: {4, 4, 2, 5, 5, 3, 2, 1, 3, 3}

➔ How do the two samples differ?

⇾ F-test to check variances, then t-Test.

⇾ There is no significant difference between the two samples.

Evaluation3: recap回顾:Observation types观察类型

➔ Direct observation直接观察⇾In the field现场(person,place,thing)⇾In controlled environments➔ Indirect observation间接观察: tracking users' activities⇾Diaries⇾Interaction logging⇾Video and photographs collected remotely by drones or other equipment

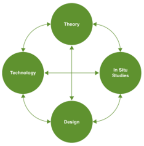
Framework框架(Robson, 2014): Space,Actors,Activities,Objects,Acts,Events,Time,Goals,Feelings

Direct observation in the field现场直接观察

➔Decide on how involved you will be⇾from passive observer to active participant从被动到积极参与➔How to gain acceptance如何获得认可➔How to handle sensitive topics处理敏感话题, for example, culture, private spaces, and so on

➔How to collect the data:收集数据⇾What data to collect⇾What equipment to use⇾When to stop observing

Research in the wild野外研究:



Observations and materials that might be collected(Crabtree2003)观察收集材料:

Activity or job descriptions/Rules and procedures that govern particular activities /Descriptions of activities observed/Recordings of the talk taking place between parties/Informal interviews with participants explaining the detail of observed activities/Diagrams of the physical layout, including the position of artifacts

Other(Photographs of artifacts (documents, diagrams, forms,computers)/Videos of artifacts/Descriptions of artifacts/Workflow diagrams showing the sequential order of tasks/Process maps showing connections between activities)

In a controlled environment在受控环境中:Direct observation直接观察:Think aloud techniques.**Indirect observation间接观察**:Tracking users'activities(Diaries⇾Interaction logs⇾Web analytics)➔ Video, audio, photos, and notes are used to capture data in both direct and indirect observations

Natural settings:Field study实地研究: Field studies are done in natural settings自然环境中"In the wild"is a term for prototypes being used freely in natural settings

➔试图了解用户自然行为Seek to understand what users do naturally and how technology impacts them

➔ Field studies are used in product design to:(⇾ Identify opportunities for new technology⇾ Determine design requirements⇾ Decide how best to introduce new technology⇾ Evaluate technology in use)

Without users没有用户:

without direct involvement of users在没有用户直接参与的情况下

➔ Experts knowledge codified in heuristics⇾Heuristic evaluation启发式评估⇾Walkthroughs演示

➔ Data collected remotely远程收集数据⇾ Analytics⇾ A/B testing

➔ Models that predict users' performance预测用户表现的模型⇾ Predictive modeling

Analytics分析:A variety of users'actions can be recorded by software automatically记录任何行为⇾Key presses⇾Mouse or other pointing device movements⇾Time spent searching a web page, looking at help systems

Advantages of logging activity automatically自动记录优势

⇾ It is unobtrusive provided the system's performance is not affected不受影响不被发现⇾Large volumes of data can be logged automatically and then explored and analyzed using visualization and other tools.记录大量数据，可视化分析

Disadvantage劣势⇾ It raises ethical concerns about observing participants if this is done without their knowledge伦理问题

Web Analytics网络分析

➔ A form of interaction logging that analyses users'activities on website互动记录

➔ Designers use the analysis to improve their designs设计师利用分析改进设计⇾When designs don't meet users' needs, they will not return to the site, they become one-time users

➔ Enable designers to track the activities of users on their site跟踪用户活动⇾How many people come to the site, how long they stay, and where they go

➔ Offer designers the "big picture" about how their site performs based on user activity

A/B Testing:

➔ A large-scale experiment (thousands of participants or more)大规模测试

➔ Offers another way to evaluate a website, application of app running on a mobile device➔ Often used for evaluating changes in design on social media applications评估设计变化➔ Compares how two groups of users perform on two versions of a design两个版本➔ May create ethical dilemmas if users don't know they are part of the test伦理困境

LMO(A/B)例子:

1.What is the independent variable?自变量

2. What is the dependent variable?因变量

3. Do you have a hypothesis?假设

4. Is it a within-subjects design or betweensubjects design?主体内还是外设计

5. How will you deal with ethical issues?道德问题

Predictive models预测模型:

➔ Provide a way of evaluating products ordesigns without directly involving users, less expensive than user testing不需要用户参与

➔ Use formulas to derive various measures of user performance公式推导

➔ Usefulness limited to systems with predictable tasks, for example, voicemail systems, smartphones, and dedicated mobile devices仅限于具有可预测任务